

Develop an elementary chatbot for any suitable customer interaction application.

```
# =====  
  
# MEDICAL CHATBOT (RULE BASED)  
  
# =====  
  
print("=====  
print("MEDICAL CHATBOT")  
print("(Educational Purpose Only)")  
print("=====  
  
# -----  
# DATABASE: Symptoms & Diseases  
# -----  
medical_db = {  
    "fever": {  
        "disease": "Viral Infection / Flu",  
        "advice": "Take rest, drink fluids, and monitor temperature.",  
        "medication": "Paracetamol (consult doctor before use)"  
    },  
  
    "cough": {  
        "disease": "Cold or Respiratory Infection",  
        "advice": "Drink warm water, avoid cold drinks, use steam inhalation.",  
        "medication": "Cough syrup (as prescribed)"  
    },  
  
    "headache": {  
        "disease": "Stress, Migraine or Fatigue",  
        "advice": "Rest in a quiet room and reduce screen time.",  
        "medication": "Pain relievers if needed"
```

},

"stomach pain": {

 "disease": "Indigestion or Gastric Problem",

 "advice": "Avoid spicy food and eat light meals.",

 "medication": "Antacids (doctor recommended)"

},

"diarrhea": {

 "disease": "Food Infection / Stomach Bug",

 "advice": "Drink ORS solution and stay hydrated.",

 "medication": "ORS and Zinc supplements"

},

"cold": {

 "disease": "Common Cold",

 "advice": "Rest, steam inhalation, warm fluids.",

 "medication": "Antihistamines (if prescribed)"

}

}

EMERGENCY SYMPTOMS

emergency_symptoms = [

 "chest pain",

 "difficulty breathing",

 "unconscious",

 "severe bleeding",

 "stroke",

 "paralysis",

```

    "seizure"
]

# -----
# FUNCTION: Show Symptoms List
# -----
def show_symptoms():

    print("\nCommon Symptoms You Can Type:")

    for symptom in medical_db.keys():
        print("-", symptom)

# -----
# FUNCTION: Emergency Check
# -----
def check_emergency(user_input):

    for symptom in emergency_symptoms:

        if symptom in user_input:
            print("\nEMERGENCY ALERT!")
            print("Possible serious condition detected.")
            print("Call ambulance immediately (India: 108 or 112)")
            return True

    return False

# -----
# FUNCTION: Diagnose
# -----

```

```

def diagnose(user_input):

    found = False

    for symptom, info in medical_db.items():

        if symptom in user_input:

            print("\n-----")
            print(f"Symptom Detected: {symptom}")
            print(f"Possible Condition: {info['disease']}")
            print(f"Advice: {info['advice']}")
            print(f"Suggested Care: {info['medication']}")
            print("-----")

            found = True

        if not found:

            print("\nSorry, I couldn't identify your symptom clearly.")
            print("Please consult a doctor for proper diagnosis.")

# -----
# MAIN CHATBOT FUNCTION
# -----
def medical_chatbot():

    print("\nHello! I am your Medical Chatbot")
    print("Type 'help' to see options or 'exit' to quit.")

    while True:

```

```

user_input = input("\nYou: ").lower()

if user_input == "exit":
    print("Stay healthy! Goodbye.")
    break

elif user_input == "help":

    print("\nOptions:")
    print("- Type your symptoms (example: fever, cough)")
    print("- Type 'list' to see symptoms")
    print("- Type 'exit' to quit")

elif user_input == "list":
    show_symptoms()

else:

    # Emergency Check
    if check_emergency(user_input):
        continue

    # Diagnose Symptoms
    diagnose(user_input)

# -----
# RUN PROGRAM
# -----
medical_chatbot()

```

This is a rule-based medical chatbot developed using Python. The chatbot interacts with the user through text input and identifies symptoms using predefined rules stored in a dictionary. It provides possible diseases, medical advice, and suggested care based on the symptom entered by the user.

The program also checks for emergency symptoms such as chest pain or breathing difficulty. If detected, it immediately displays emergency instructions. The chatbot uses functions, loops, conditional statements, and dictionaries to simulate basic medical assistance.